

Spring 2026 Data C100/C200 Midterm 3 Reference Sheet

Probability

Let X have a discrete probability distribution $P(X = x)$. X has expectation $\mathbb{E}[X] = \sum_x xP(X = x)$ over all possible values x , variance $\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2]$, and standard deviation $\text{SD}(X) = \sqrt{\text{Var}(X)}$.

Notes	Property of Expectation	Property of Variance
X is a random variable.	$\mathbb{E}[X] = \sum_x xP(X = x)$	$\text{Var}(X) = \mathbb{E}[(X - \mathbb{E}[X])^2] = E[X^2] - (E[X])^2$
X is a random variable, $a, b \in \mathbb{R}$ are scalars.	$\mathbb{E}[aX + b] = a\mathbb{E}[X] + b$	$\text{Var}(aX + b) = a^2\text{Var}(X)$
X, Y are random variables.	$\mathbb{E}[X + Y] = \mathbb{E}[X] + \mathbb{E}[Y]$	$\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{Cov}(X, Y)$
X is a Bernoulli random variable that takes the value 1 with probability p , and 0 otherwise.	$\mathbb{E}[X] = p$	$\text{Var}(X) = p(1 - p)$

Parameter Estimation

Suppose for each individual with fixed input x , we observe a random response $Y = g(x) + \epsilon$, where g is the true relationship and ϵ is random noise with zero mean and variance σ^2 .

For a new individual with fixed input x , define our random prediction $\hat{Y}(x)$ based on a model fit to our observed sample $(\mathbb{X}, \mathbb{Y})$. The model risk is the mean squared prediction error between Y and $\hat{Y}(x)$: $\mathbb{E}[(Y - \hat{Y}(x))^2] = \sigma^2 + \left(\mathbb{E}[\hat{Y}(x)] - g(x)\right)^2 + \text{Var}(\hat{Y}(x))$.

Suppose that input x has p features and the true relationship g is linear with parameter $\theta \in \mathbb{R}^{p+1}$. Then $Y = f(x) = \theta_0 + \sum_{j=1}^p \theta_j x_j + \epsilon$ and $\hat{Y} = \hat{f}(x)$ for an estimate $\hat{\theta}$ fit to the observed sample $(\mathbb{X}, \mathbb{Y})$.

SQL

SQL syntax:

```
SELECT [DISTINCT]
      { * | expr [[AS] c_alias]
        {, expr [[AS] c_alias] ...} }
FROM tableref {, tableref}
[[INNER | LEFT | RIGHT | FULL | CROSS] JOIN table_name
  ON qualification_list]
[WHERE search_condition]
[GROUP BY colname {, colname...}]
[HAVING search_condition]
[ORDER BY column_list]
[LIMIT number]
[OFFSET number of rows];
```

Strings in SQL should use single quotes. Column names that contain a space should use double quotes.

Syntax	Description
<code>SELECT column_expression_list</code>	List is comma-separated. Column expressions may include aggregation functions (<code>MAX</code> , <code>FIRST</code> , <code>COUNT</code> , <code>AVG</code> , etc). <code>AS</code> renames columns. <code>DISTINCT</code> selects only unique rows.
<code>FROM s INNER JOIN t ON cond</code>	Inner join tables <code>s</code> and <code>t</code> using <code>cond</code> to filter rows; the <code>INNER</code> keyword is optional.
<code>FROM s LEFT JOIN t ON cond</code>	Left outer join of tables <code>s</code> and <code>t</code> using <code>cond</code> to filter rows.
<code>FROM s RIGHT JOIN t ON cond</code>	Right outer join of tables <code>s</code> and <code>t</code> using <code>cond</code> to filter rows.
<code>FROM s FULL JOIN t ON cond</code>	Full outer join of tables <code>s</code> and <code>t</code> .
<code>FROM s CROSS JOIN t</code>	Explicit cross join of tables <code>s</code> and <code>t</code> ; Equivalent to <code>FROM s, t</code> .
<code>FROM s, t</code>	Cross join of tables <code>s</code> and <code>t</code> : all pairs of a row from <code>s</code> and a row from <code>t</code>

Syntax	Description
<code>WHERE a IN cons_list</code>	Select rows for which the value in column <code>a</code> is among the values in a <code>cons_list</code> .
<code>ORDER BY RANDOM() LIMIT n</code>	Draw a simple random sample of <code>n</code> rows.
<code>ORDER BY a, b DESC</code>	Order by column <code>a</code> (ascending by default) , then <code>b</code> (descending).
<code>CASE WHEN pred THEN cons ELSE alt END</code>	Evaluates to <code>cons</code> if <code>pred</code> is true and <code>alt</code> otherwise. Multiple <code>WHEN/THEN</code> pairs can be included, and <code>ELSE</code> is optional.
<code>WHERE s.a LIKE 'p'</code>	Matches each entry in the column <code>a</code> of table <code>s</code> to the text pattern <code>p</code> . The wildcard <code>%</code> matches at least zero characters.
<code>LIMIT number</code>	Keep only the first <code>number</code> rows in the return result.
<code>OFFSET number</code>	Skip the first <code>number</code> rows in the return result.

Logistic Regression and Classification

Logistic Regression Model: For input feature vector x , $\hat{P}_\theta(Y = 1|x) = \sigma(x^T\theta)$, where $\sigma(z) = 1/(1 + e^{-z})$. For a single datapoint, define cross-entropy loss as $-[y\log(p) + (1 - y)\log(1 - p)]$, where p is the probability that the response is 1.

An ROC curve has the false positive rate (FPR) on the x-axis and true positive rate (TPR) on the y-axis.

Classification Performance

Suppose you predict n datapoints.

Metric	Formula
Accuracy	$\frac{TP+TN}{n}$
Precision	$\frac{TP}{TP+FP}$
Recall, True Positive Rate (TPR)	$\frac{TP}{TP+FN}$
False Positive Rate (FPR)	$\frac{FP}{FP+TN}$
F1 Score	$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$

Clustering

A datapoint's **silhouette score** S is defined as $S = (B - A)/\max(A, B)$, where A is the mean distance to other points in its cluster, and B is the mean distance to points in its closest cluster.